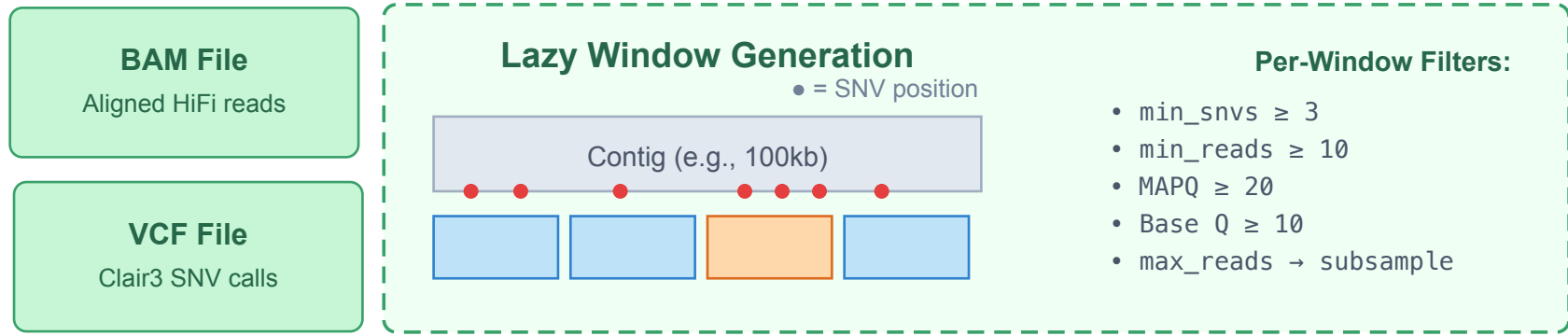


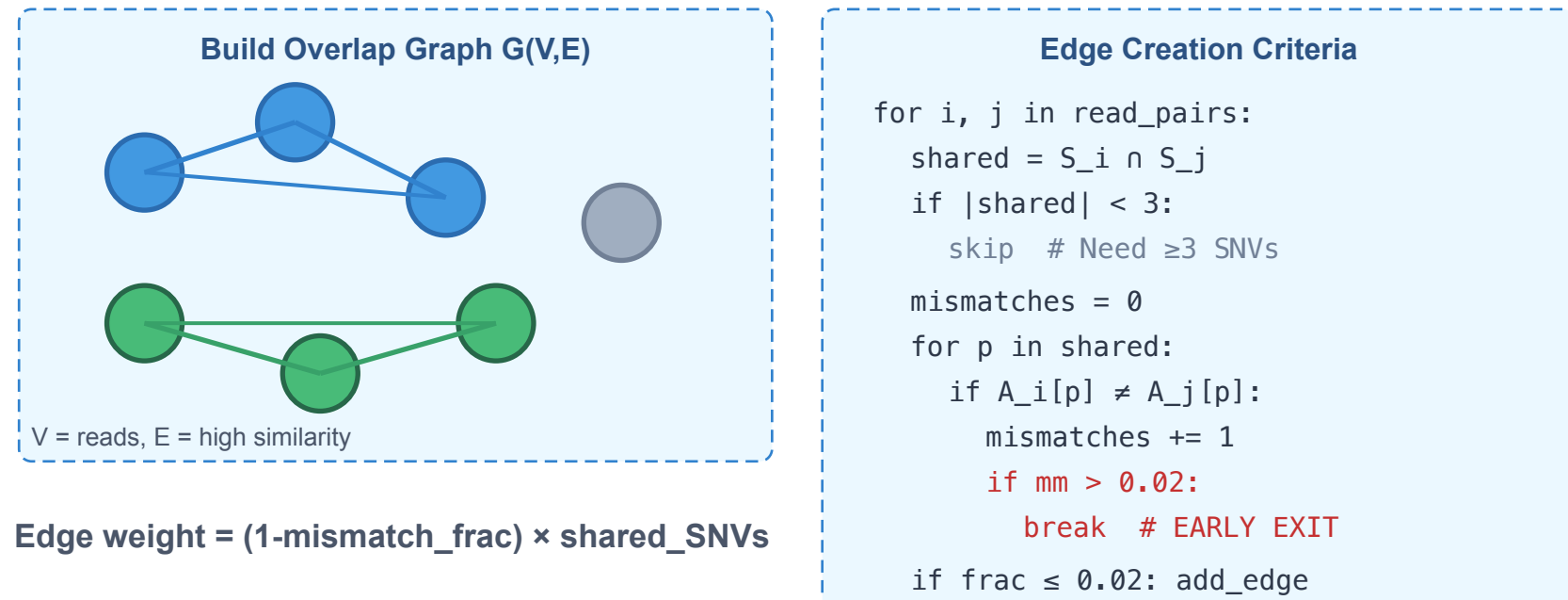
Hybrid Graph-Probabilistic Haplotype Reconstruction

for PacBio HiFi Metagenomic Data

1. INPUT & WINDOWING

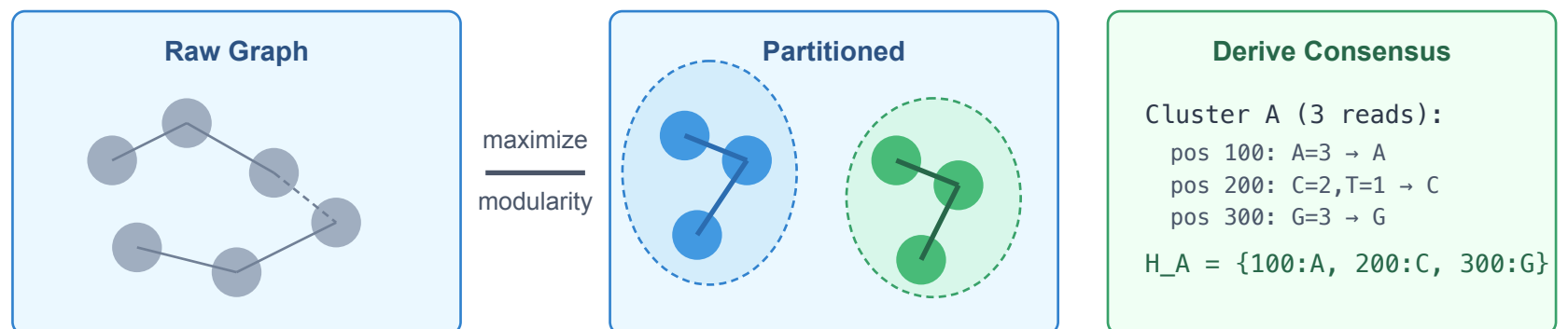


2. GRAPH INITIALIZATION



3. LOUVAIN COMMUNITY DETECTION

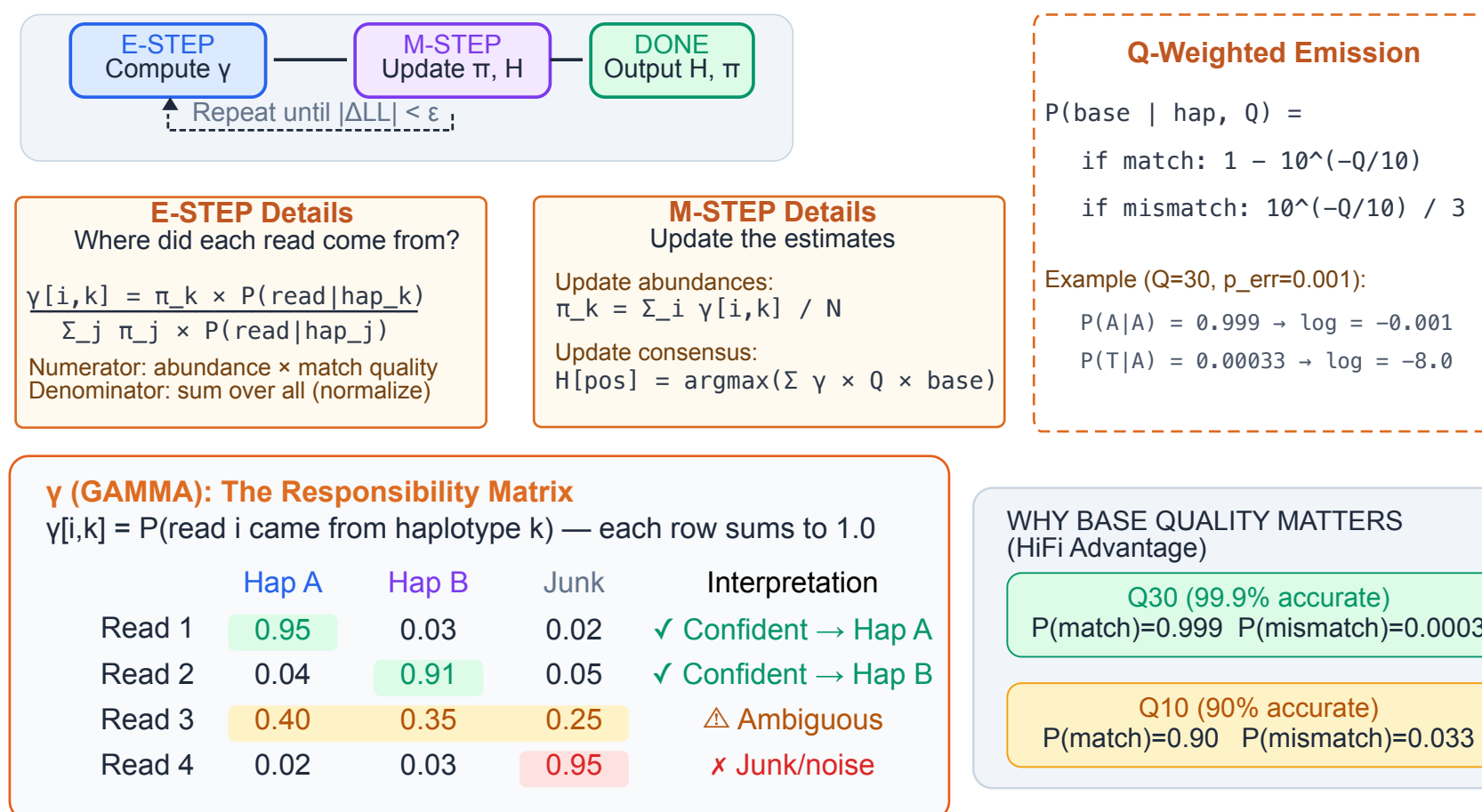
Iteratively move nodes to neighboring communities if it increases modularity



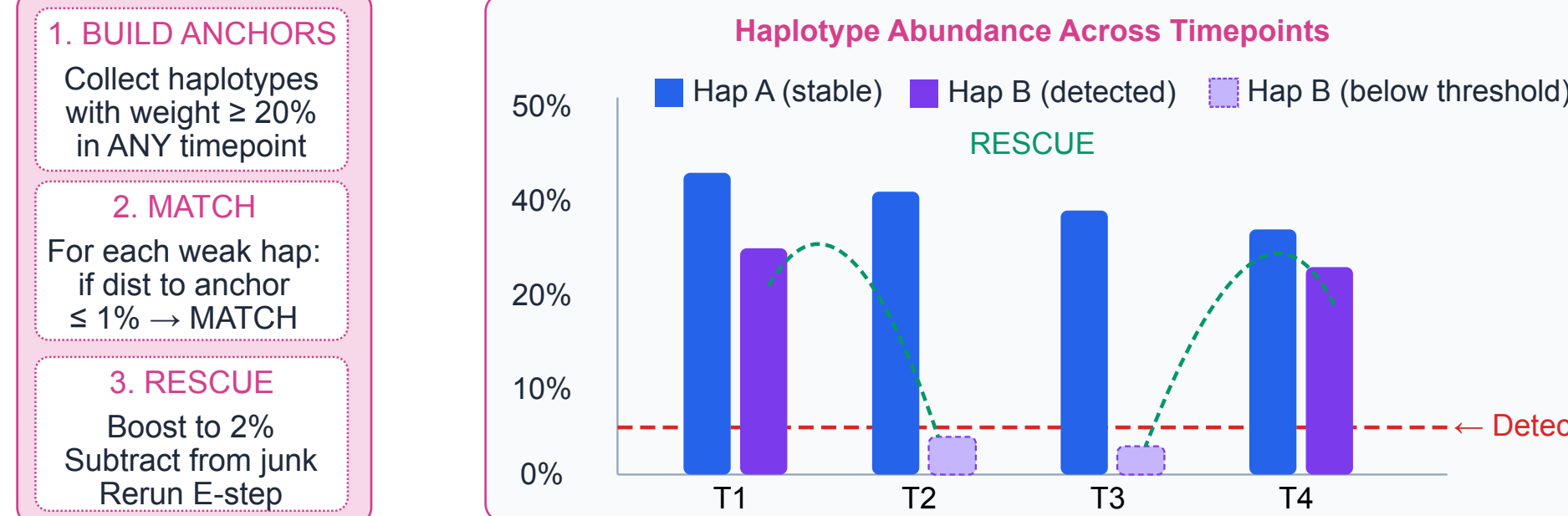
5. POST-PROCESSING & MERGING



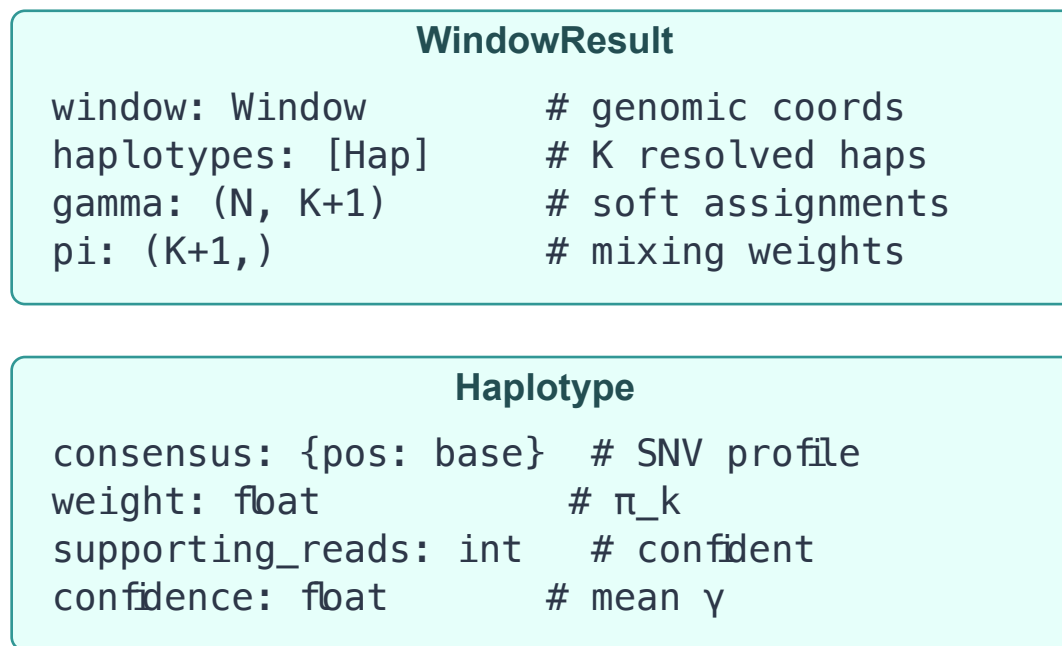
4. EXPECTATION-MAXIMIZATION REFINEMENT



6. LONGITUDINAL INTEGRATION (Multi-Timepoint)



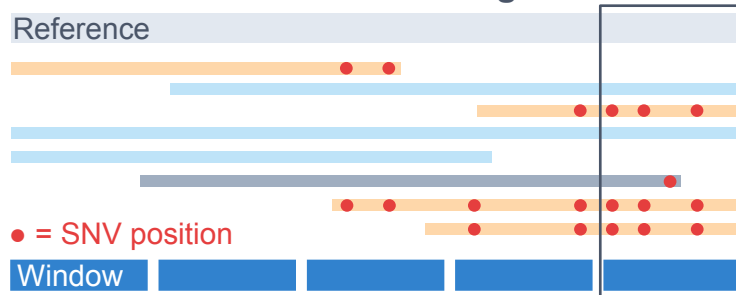
7. OUTPUT STRUCTURE



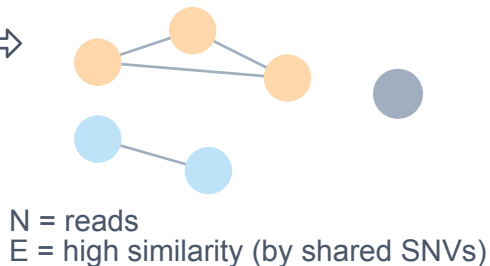
Configuration Parameters (HaplotyperConfig)

Window & Filtering window_size = 3000 min_snvs_per_window = 3 min_reads_per_window = 10 max_reads_per_window = 300 min_mapq = 20 min_base_quality = 10	Graph Construction min_shared_snvs_for_edge = 3 max_mismatch_frac = 0.02 min_reads_per_cluster = 3 EM Engine em_max_iter = 20 em_tolerance = 1e-4 min_hap_eff_weight = 3.0	Junk Model junk_divergence_rate = 0.10 Post-Processing merge_distance_threshold = 0.01 assign_confidence_threshold = 0.90 1-SNP Validation min_minor_frequency_1snp = 0.10 min_timepoints_for_1snp = 2	Longitudinal Rescue min_weight_for_anchor = 0.20 rescue_match_distance = 0.01 rescued_min_weight = 0.02 Runtime random_seed = None validate_results = True	SNV Filtering min_depth_site = 10 af_range = (0.01, 0.99) require_biallelic = True skip_af_filter_if_missing = True
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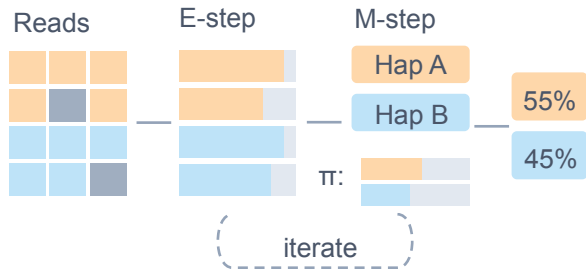
Generate Windows from Long Reads



Build Overlap Graph $G(V,E)$



EM Refinement



Longitudinal Integration + Rescue

